

Actuator HW4

May 7, 2026

A DC motor with serial excitation has the following specifications:

$$\begin{aligned} V_n &= 220 [V] & R_a + R_f &= 0.3 [\Omega] & P_n &= 10 [kW] \\ n_n &= 1800 [rpm] & \eta_n &= 88\% \end{aligned}$$

The motor has negligible brush losses, has a constant torque loss, and $k_e = k_t$ (in SI units).

- a. The torque applied to the motor (not at the nominal working point) is increased such that the speed falls to 1000 rpm. Find the new torque on the motor, and the efficiency at this torque.

Hint: P_n is the power output of the motor, which is $P_n = (k_t I - \tau_{\text{const}})\omega_n$, where I is the current on the armature, τ_{const} is the constant torque loss.

$$P_{in} = \frac{P_n}{\eta_n} = 1111363.6W$$

$$P_{in} = V_{in} I \Rightarrow I = \frac{P_{in}}{V_n} = \frac{11363.6}{220} = 51.65[A]$$

$$V_n = I(R_A + R_f) + E_A \Rightarrow E_A = 204.505[V]$$

$$E_A = K\omega = kk_\phi I\omega = k_e\omega$$

where $K = k\phi = kk_\phi I$, $\omega = \frac{1800 \cdot 2\pi}{60} = 188.5[rad/s]$. thus $kk_\phi = 0.021$, $k_e = 0.021 \cdot I$

$$P_n = k_t I - \tau_{\text{const}}\omega_n$$

$$10000 = (0.021 \cdot I^2 - \tau_{\text{const}} \cdot 188.5) \Rightarrow \tau_{\text{const}} = 2.98[N \cdot m]$$

When $n' = 1000rpm$, $\omega'_n = 104.72 rad/s$

$$220 = E'_A + I'(R_A + R_f) = 0.021 \cdot I'^2 \omega'_n + I' \cdot 0.3 \Rightarrow I' = 88.04A$$

$$T'_{out} = T - \tau_{\text{const}} = kk_\phi I'^2 - 2.98 = 162.77 - 2.98 = 159.8[N \cdot m]$$

$$P'_n = (kk_\phi I'^2 - \tau_{\text{const}})\omega'_n = (0.021 \cdot 88.04^2 - 2.98) \cdot 104.72 = 16733.405W$$

$$P'_{in} = V I' = 220 \cdot 88.04 = 19368.8W$$

$$\eta' = \frac{P'_n}{P'_{in}} = \frac{16733.405}{19368.8} \approx 86.4\%$$

During the project you built a permanent magnet DC motor and conducted the following tests:

- I. No load test at $V_1 = 10[V]$ yielded the no load velocity $\omega_{0,1} = 21 [rad/s]$.
- II. Stall test at $V_1 = 10[V]$ yielded the stall torque $T_{st} = 11/48 [Nm]$.
- III. No load test at $V_2 = 12.5[V]$ yielded the no load velocity $\omega_{0,2} = 27 [rad/s]$.

Assuming negligible brush voltage drop ($\Delta U_{br} = 0$) and constant torque losses, calculate:

- a. The motor constant K (where $K = k\Phi$)
- b. The rotor resistance R_a
2. c. The mechanical torque losses ΔT

a)

According to KVL: $V = IR_A + E_A + 2\Delta U_{br}$, here $\Delta U_{br} = 0$

$$V = IR_A + E_A, T = T_{load} + \Delta T$$

$$V = \frac{T}{K} + K\omega$$

where $I = \frac{T}{K}$

Case I and III:

$$T_{load} = 0, T = \Delta T$$

$$V_1 = \frac{\Delta T}{K} R_A + K\omega_1$$

$$V_2 = \frac{\Delta T}{K} R_A + K\omega_2$$

$$V_1 - V_2 = K(\omega_1 - \omega_2)$$

$$-2.5 = -6K \Rightarrow K = \frac{5}{12} \approx 0.4167$$

b) Case II

$$V_1 = I_{st} R_A = \frac{T_{st}}{K} R_A \Rightarrow R_A = 18.2[\Omega]$$

c)

$$V_1 = \frac{\Delta T}{K} R_A + \omega_1$$

$$\Delta T = \frac{(V_1 - \omega_1 K) K}{R_A} = 0.0286 [N \cdot m]$$